ktwugoat@gmail.com
ktwu01 ktwu01

Koutian Wu

(+1) 737-420-8027
Flexible with relocation

Seeking internships in AI Agent Development/Test/Evaluation, ML/DL/DS, Full-Stack Engineering
2+ years work experience (U.S./International); **4+** years research/programming experience; **302** stars on GitHub.

Education

Ph.D. Student in AI for Geosciences | **University of Texas at Austin** | 2024 – Expected: May 2029

TA for 1,000+ students (Rating: 4.5/5.0)
Reviewer for *ACM Computing Surveys* (JCR Q1), *ACM AgentSkills*, and *Planetary and Space Science*

B.S. in Geosciences (Honors Class) | **University of Science and Technology of China** | 2020 – 2024

Finalist for the Guo Moruo Scholarship (USTC's highest undergraduate honor)
Founded one of the largest student clubs, grew from 0 to 1,300+ members. [News]

Work Experience

CEO-Invited AI Consultant | **Keshi Edu** | Remote, Beijing | May 2025 – Jun. 2025

- Designed an architecture for a RAG-based knowledge base to customize graduate program applications.
- Identified critical bugs and exposed API keys, saving thousands of dollars in potential losses.

AI Intern, AI Agent Knowledge Base Evaluation | **19Pine.AI** (Singapore) | Remote | Jul. 2025 – Aug. 2025

- Implemented a multi-dimensional evaluation system from scratch for PineAgent's RAG knowledge base.
- Developed a data sanitization module removing 2,000+ PII entries and noise from 1,000+ call sessions.
- Extracted Q&A pairs (knowledge/method/strategy) from call sessions to form the evaluation dataset.
- Deployed a concurrent LLM-as-judge + rule-based engine, measuring P/R/F1 to enable RLHF optimization.

Full-Stack Intern, LLM Text Processing System | **ZaiwenAI.com** | Beijing, China | Jun. 2025 – Jul. 2025

- Developed a 3-module NLP system (LLM-text detection, paraphrasing, plagiarism checking) with FastAPI + Celery + Redis backend, supporting 9 document formats and SSE-based real-time LLM streaming.

Lead Developer, ESM-bench: AI Agent Benchmark for Earth System Models | UT-Austin | 2026 – Present

- Designed a 243-task benchmark across 3 Earth System Models (Noah-MP, CLM5, MOM6) evaluating whether LLM agents can bridge the physics-and-code understanding gap in 200k+ line Fortran/C codebases.
- Built a multi-model evaluation framework testing 5+ LLMs (GPT-5.5, Claude Opus, Gemini, etc) with 3 progressive hint levels and a 6-item physics-aware semantic and result-based rubric.
- Achieved <20% exact match across all frontier models, demonstrating that physical correctness in scientific code remains an unsolved challenge for current LLMs. [Preprint]

Research Assistant, ML Model/Multi-agent Development | **UT-Austin** | Austin, TX | Aug. 2025 – Jan. 2026
Project Lead, High-Performance Computing Allocation | **NSF NCAR** | Remote | Aug. 2025 – Present

- Proposed and secured NSF NCAR's 1k GPU + 22k CPU hours high-performance computing allocation.
- Integrating machine learning parameter calibration for a SOTA physics-based land surface model (Noah-MP).
- Developing a multi-expert AI agent system for automated parameterization for physics-based climate models.

Publications

- **Wu, K.**, Cao, Y., & Mai, G. (2026). ESM-bench: A Benchmark for Evaluating Whether AI Agents Understand Earth System Model Physics and Code. (In prep to NeurIPS '26 dataset). [Preprint]
- **Wu, K.**, & Huang, Z. (2026). Commit Log as Skill Mine: Extracting 16 Agent Skills from a Scientific Research Trajectory. (Under Review for ACM AgentSkills 2026 Workshop). [Preprint]
- **Wu, K.**. Noah-Agent: A Multi-Expert AI Agent Framework for Automated Parameterization and Validation of Large-Scale Fortran Climate Models. (In prep). [Record]
- **Wu, K.**, Yi, W.*, Xue, X.*, Reid, I., & Lu, M. (2024). Diurnal and Seasonal Variations of Meteor Speed and Arrival Angle Observed by Mengcheng Meteor Radar. *JGR: Space Physics*. [Paper] [Code]

Skills

- Python, FastAPI, JavaScript, TypeScript, React, SQL, PyTorch, TensorFlow, LangChain, RAG
- Docker, HPC, AWS/GCP, Redis, Git, HuggingFace